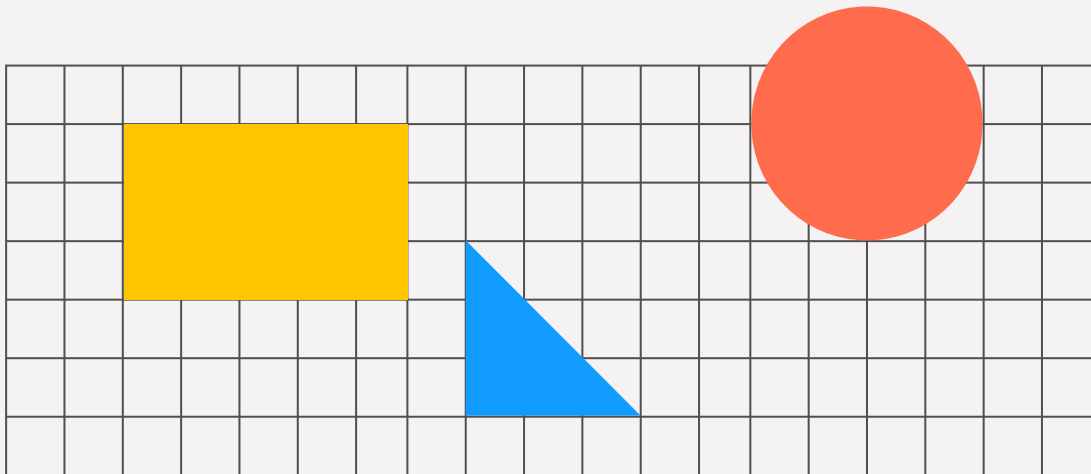


Session 1, Sep. 18, 2025

Getting Started with Research Data: Concepts, Tools, & Resources

Wei Yin | wy2288@columbia.edu
Research Support & Data Services Librarian
[Research Data Services](#), Columbia University Libraries



Research Data Services (RDS) helps researchers across Columbia find, evaluate, understand, steward, and use a range of statistical, geospatial, and textual data. Our expert staff provide training and consultation around a number of tools for analyzing, visualizing, and managing data.

Data Club welcomes members of the Columbia community who work with data and are interested to learn more about data analysis, management, and other aspects of the data life cycle at our meetings.

We offer a supportive meeting space for members of the Columbia community of all levels (beginners and experts alike!) to learn, share, and implement the philosophies, tools, and methodologies for a data-driven project of their choosing.

RDS Email

data@library.columbia.edu

Join Data Club

Data Club Linktree

Where we meet

215 **Lehman Library**

Who we are



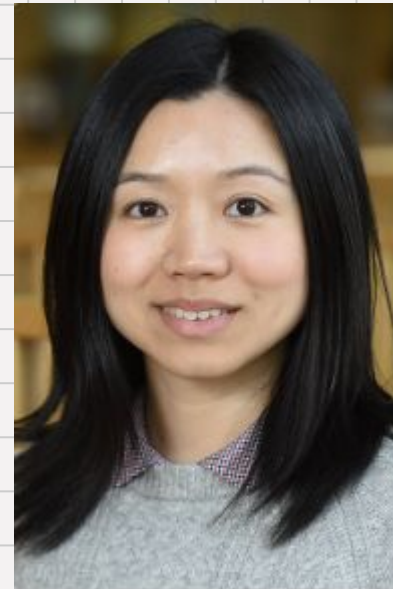
Eric



Ask Moacir for more
information about
Data Club!



Jeremiah



Wei

Schedule

Date	Topic	Facilitator
Thursday Sep 18	Getting Started with Research Data: Concepts, Tools, & Resources	• Wei Yin • Research Support & Data Services Librarian • wy2288@columbia.edu
Thursday Sep 25	Introduction to Open Source GIS	• Eric Glass • GIS/Metadata Librarian • ecg2104@columbia.edu
Thursday Oct 2	Introduction to Python and Text Mining with TDM Studio and ChatGPT	• Moacir P. de Sá Pereira • Research Data Librarian • mpd2149@columbia.edu
Thursday Oct 9	Working with Large Datasets in R	• Wei Yin • Research Support & Data Services Librarian • wy2288@columbia.edu

Schedule

Date	Topic	Facilitator
Thursday Oct 23	Data Engineering with Pandas and Polars in Python	• Moacir P. de Sá Pereira • Research Data Librarian • mpd2149@columbia.edu
Thursday Oct 30	Using American Indian and Alaska Native Census Data	• Jeremiah Trinidad-Christensen • Head, Research Data Services • jt2118@columbia.edu
Thursday Nov 6	Open Source Geospatial Data Formats	• Eric Glass • GIS/Metadata Librarian • ecg2104@columbia.edu
Thursday Nov 13	Getting Access to Restricted Health Care Utilization Project Data	• Jeremiah Trinidad-Christensen • Head, Research Data Services • jt2118@columbia.edu

Data Formats

Perspective 1



Name	Age	GPA
Alex	22	3.8

Structured

- Hierarchical, relational and tabular
- Such as spreadsheets, parquet, GeoPackage, NetCDF, FITS, FASTA, etc.



```
{  
  "students": [  
    {  
      "Name": "Alex",  
      "Age": 22,  
      "GPA": 3.8  
    }  
  ]  
}
```

Semi-structured

- Nested hierarchy
- Such as JSON, XML, HTML, etc.



Alex is a 22-year-old
student with a GPA of
3.8

Unstructured

Such as Text, Sensor, Video, Audio,
Multimedia

Using Geospatial Data as an example

- Talk with Eric if you have any further questions related to Geospatial data
- Attend Eric's sessions
 - [Open source GIS \(Sep. 25\)](#)
 - [Geospatial Data Formats \(Nov. 6\)](#)

Structured

Such as shapefile, GeoPackage, PostGIS DB, GeoTIFF (w/metadata)

Semi-Structured

Such as GeoJSON, KML, GML

unstructured

Such as raw images, scanned maps, field notes

Data Types

Perspective 2

- [NYC Open Data](#)
- [OpenICPSR](#)
- [OpenStreetMap](#)
- [Columbia's Academic Commons](#)

Fully Open Data

public and free to use with no restrictions

- Jeremiah's session about [American Indian and Alaska Native Census Data](#) (Oct. 30)
- Generally speaking, US Census data fall into all the three types of data

Conditional Use Data

- Moacir's session about [text mining using TDMStudio](#) (Oct. 2)
- Jeremiah's session about [getting access to HCUP data](#) (Nov. 13)

Restricted Data

with confidential, sensitive, or proprietary information

Popular Data Terms in Research

Expand your understanding of data intensive research



- Data simulation
- Synthetic data
- Big data
 - How big is big?
 - How to deal with it?
 - What discipline(s) are you interested in?
- Artificial Intelligence /Machine Learning /Deep Learning



- Moacir's [Python session about pandas and polars](#) from a data engineering perspective (Oct. 2)
- Wei's [R session about working with large datasets](#) from a data analytics perspective (Oct. 9)

Popular Data Tools in Research

Expand your understanding of data intensive research

- Statistical analysis: R, Stata, SPSS, SAS, etc.
- ML/DL (more towards large-scale computing): Python
- SQL
- GIS
- [Qualitative data tools](#)

Find more learning resources on the next slide!

- Excel / Google sheets
- Tableau / PowerBI
- OpenRefine
- JavaScript / Observable

Although this semester's data club won't cover these tools, you can reach out to us for further conversations.

Research Data / Computing Resources on Campus

RDS Libguides

- [Data Analysis Tools](#)
- [GIS & Geospatial Data](#)
- [GIS, Cartographic, & Spatial Analysis Tools](#)
- [Text Mining](#)
- [Health Data](#)

And more libguides [here](#) (lower right corner)

CUIT

- [High Performance Computing](#)
- [AI Services](#)
- Contact [Research Computing Services](#) and check for more information on its [research services webpage](#)

Before you leave this session

A few suggestions for building up and strengthening your data skills

- At least learn one type of data tool and/or programming language
- Stick with your coursework and research project
- If you feel uncomfortable or anxious about learning data skills, finding data sources and beyond, contact us!
- Do you have any questions now? Ask us!

data@library.columbia.edu

[Data Club Linktree](#)

**Just
Remember !**